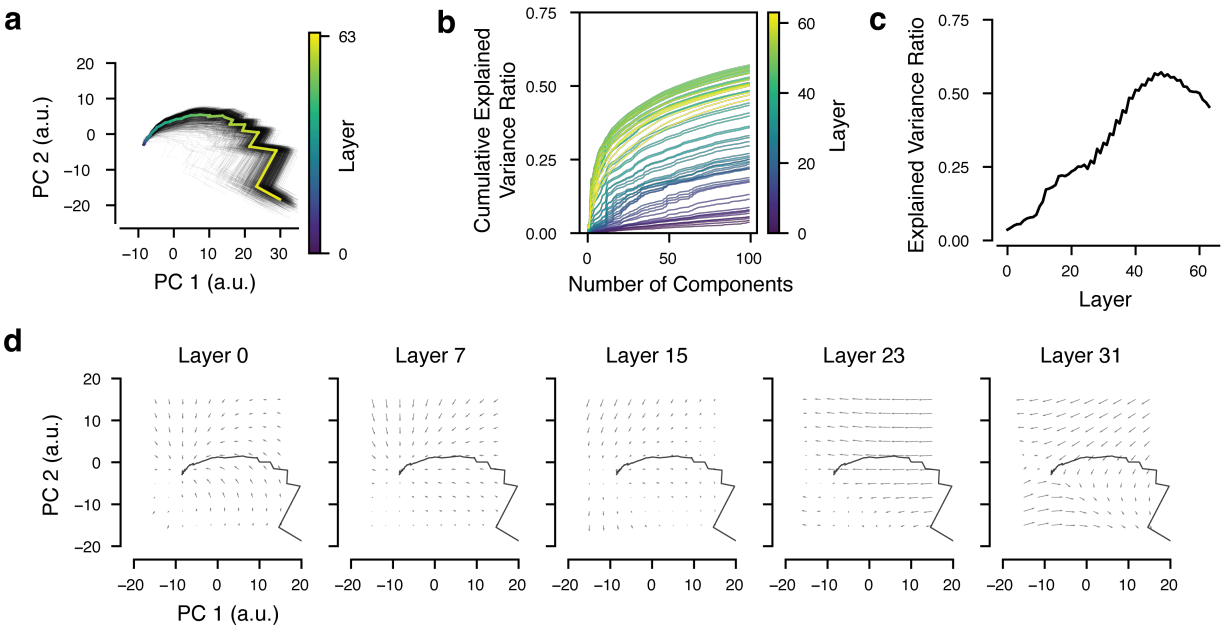

Figure 4. Perturbation of RS trajectories Reveals Self-correcting Dynamics **A:** Trajectories of $n = 1000$ individual (black) and mean (colored by layer) data samples in PCA space. **B:** Cumulative explained variance of the trajectories as a function of the number of components. **C:** Explained variance per layer using 100 PC components. **D:** Perturbation analysis in which trajectories were 'teleported' to various points, at various stages in the RS (indicated by layer number above each subplot). Gray line shows unperturbed control trajectory. Quiver arrows indicate direction and magnitude of teleported trajectories based on the successive 12 sublayers after teleportation.

in this phase space were not particularly smooth, with abrupt changes in direction from one sublayer to the next, the overall portraits showed clear and consistent rotational patterns, with a mean of 10.74 rotations over the layers compared to approximately zero rotations in shuffle controls (Fig. 2B). These rotations often spiraled out, increasing in magnitude on both axes as they progressed through the layers. Further, the rotations were often centered at multiple points in the phase space, not only at the origin.

This rotational behavior was consistent across the majority of the 4096 units in the RS (Fig. 2C), suggesting a systematic organizational principle in the network's computation.

These observations are consistent with the data on increased velocity of the RS as a whole, and illustrate that individual RS units exhibit complex dynamics rather than simply increasing their activation magnitudes linearly.

3.3. Reduced Dimensional Trajectories

To analyze the high-dimensional activation patterns in the RS, we used complementary dimensionality reduction approaches: a compressing autoencoder (CAE) (Fig 3) and PCA (Fig 4). The CAE was trained to pass RS vectors through successively lower-dimensional layers, until passing through a 2-dimensional bottleneck and being reconstructed again. Here RS vectors at each sublayer were treated as individual samples to be learned. The bottleneck representation revealed structured, curving trajectories (Fig. 3B). The trajectories in this space straightened over the layers, with increasing distance between successive layers (Fig. 3C). The model's reconstruction of RS vectors showed increasing explained variance, with a large jump after the early layers and a slower increase thereafter (Fig. 3D).

Principal Component Analysis revealed that early layers distribute variance across more dimensions than later layers, with later layers requiring fewer principal components to explain the same amount of variance (Fig. 4A,B,C).

3.4. RS Trajectories Exhibit Attractor-like Dynamics in Lower Layers

Visualization of RS trajectories in PCA space revealed systematic patterns in how representations evolve through the network (Fig. 4A). Individual trajectories and their layer-wise means demonstrated a consistent path through this reduced space, suggesting a structured computation process, with slightly offset trajectories for the within-layer $\mathbf{h}_l^{Attn} \rightarrow \mathbf{h}_l^{MLP}$ and cross-layer $\mathbf{h}_l^{MLP} \rightarrow \mathbf{h}_{l+1}^{Attn}$ transitions.

To understand the stability of these trajectories, we performed perturbation analysis by "teleporting" the RS state to various points in the PCA space at different layers (Fig.

4D). We hypothesized that RS progression through transformer layers might reveal attractor-like dynamics, such that moving the activations to various portions of this phase space would eventually bring them back to the mean trajectory.

The response to these perturbations varied systematically with layer depth and location of the teleported points. For instance,

Still, the effect of the perturbations varied systematically. Perturbations at layer 0 (i.e. starting the trajectories at new points) resulted in highly variable dynamics, with trajectories often ending up far from the unperturbed mean. This effect was even more drastic when interfering in the dynamics at the penultimate LLM layer, 31. Mid-layer perturbations (layers 7, 15, 23) exhibited a more robust recovery, with lower variance on the perturbed trajectories and lower mean squared error between the control sequence (Fig. 4D).

This data suggests that the transformer develops stable computational channels that actively maintain desired trajectories, possibly self-correcting errors through its dynamics.

4. Discussion

In this work, we approached mechanistic interpretability of transformer circuits through a dynamical systems lens inspired by neuroscience. Treating the residual stream (RS) of the Llama 3.1 8B LLM, we found that individual RS units exhibited increasingly strong correlations from layer to layer while growing in magnitude but with a sharp drop in mutual information after the early layers. RS units displayed rotational structures, with dynamics reminiscent of unstable periodic orbits. The RS vector as a whole increased in density of activations, with more alignment of the vector at successive layers as revealed by cosine similarity. Dimensionality reduction methods revealed low-dimensional dynamics as a whole, with trajectories that curved in low-dimensional space before straightening out and jumping progressively farther at each successive layer. Finally, perturbations to the RS at various layers revealed a self-correcting tendency, returning immediately close by to original spots in reduced space, akin to a pseudo-attractor.

In the broader context of mechanistic interpretability, whereas recent advances have focused on circuits (Singh et al., 2024; Kissane et al., 2024) or sparse autoencoders (Bricken et al., 2023), the dynamical systems approach to understanding transformers is nascent and has largely received theoretical treatment. Investigating the dynamics of complicated systems such as transformers could help unify theoretical insights from dynamical systems theory with large-scale data analysis and experimental manipulation.

While the present results are based on a popular open source model, Llama 3.1, preliminary analysis on another model, Gemma 2, showed similar results (data not shown). Future work beyond LLMs may examine dynamics of the residual stream in other AI architectures with a residual stream, i.e. ResNets. Moreover, while we presently focused on encoded sequences only at the last token position, it will be crucial to understand how whole sequences of tokens influence dynamics. The observed dynamics likely evolved over the course of model training, and subsequent work may investigate such dynamics over the course of learning.

Despite the high dimensionality of the residual stream (D=4096), the dynamics we observe are remarkably low-dimensional, as shown by both the autoencoder bottleneck and the interpretable structure in PCA space. This low-dimensional behavior may reflect the relative simplicity of our experimental task of encoding high-probability sequences of tokens from Wikitext. More complex tasks, like those requiring in-context learning or complex reasoning, may reveal richer dynamical structures by taking advantage of the network's representational capacity. This suggests future work examining how the dimensionality and structure of these dynamics scales with task complexity.

Finally, while perturbing activations by 'teleporting' them to various portions of reduced-dimensional space was revealing, it is possible this approach does not realistically capture how a model might respond to more subtle changes to its activations. Future efforts may attempt noise injection or swapping of activations from one input data sample to another.

The discovery of rotational dynamics in activation-gradient space, combined with the self-correcting properties observed in our perturbation analysis, points to an emerging organizational principle. The network appears to construct stable computational channels that actively maintain desired trajectories. This self-correction is most robust in lower layers, where cosine similarity and velocity among succeeding RS vectors are lowest, and mutual information the highest. Finally, the strong correlations and low-dimensional flows imply that the network may perform highly distributed computations rather than localizable "grandmother cell" style encoding. Insights such as presented here could inform both a theoretical understanding of transformer dynamics and practical approaches to architecture design and training optimization.

Impact Statement

This work explores the mechanistic interpretability of transformers from a dynamical systems perspective inspired by neuroscience. By bridging dynamical systems theory with transformer interpretability, we introduced a novel way to understand the behavior of LLMs. This approach follows

information flows and transformations through neural networks, similar to how neural computations in the brain evolve over time. Our findings about residual stream dynamics and their self-correcting properties can inform the development of more reliable and efficient AI systems. This interdisciplinary approach unlocks a better understanding of both biological and artificial systems, and allows AI researchers to build more interpretable systems. This work contributes to the larger goal of creating AI systems that are transparent and understandable for safe and ethical deployment in society.

References

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., and Mané, D. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.

Bahri, Y., Dyer, E., Kaplan, J., Lee, J., and Sharma, U. Explaining neural scaling laws. *Proceedings of the National Academy of Sciences*, 121(27):e2311878121, 2024.

Barack, D. L. and Krakauer, J. W. Two views on the cognitive brain. *Nature Reviews Neuroscience*, 22(6):359–371, June 2021. ISSN 1471-003X, 1471-0048. doi: 10.1038/s41583-021-00448-6. URL https://www.nature.com/articles/s41583-021-00448-6.

Bengio, Y., Hinton, G., Yao, A., Song, D., Abbeel, P., Darrell, T., Harari, Y. N., Zhang, Y.-Q., Xue, L., Shalev-Shwartz, S., and others. Managing extreme AI risks amid rapid progress. *Science*, 384(6698):842–845, 2024. Publisher: American Association for the Advancement of Science.

Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., and others. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2, 2023.

Geshkovski, B., Letrouit, C., Polyanskiy, Y., and Rigollet, P. A mathematical perspective on Transformers, August 2024. URL http://arxiv.org/abs/2312.10794. arXiv:2312.10794 [cs, math].

Heimersheim, S. and Nanda, N. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

Hosseini, E. A. and Fedorenko, E. Large language models implicitly learn to straighten neural sentence trajectories to construct a predictive representation of natural language, November 2023. URL http://arxiv.org/abs/2311.04930. arXiv:2311.04930 [cs].

Inagaki, H. K., Fontolan, L., Romani, S., and Svoboda, K. Discrete attractor dynamics underlies persistent activity in the frontal cortex. *Nature*, 566(7743):212–217, 2019.

Kissane, C., Krzyzanowski, R., Bloom, J. I., Conmy, A., and Nanda, N. Interpreting Attention Layer Outputs with Sparse Autoencoders, June 2024. URL http://arxiv.org/abs/2406.17759. arXiv:2406.17759 [cs].

Lawson, T., Farnik, L., Houghton, C., and Aitchison, L. Residual Stream Analysis with Multi-Layer SAEs, October 2024. URL http://arxiv.org/abs/2409.04185. arXiv:2409.04185 [cs].

Lorenz, E. N. Deterministic nonperiodic flow. *Journal of atmospheric sciences*, 20(2):130–141, 1963.

Lu, Y., Li, Z., He, D., Sun, Z., Dong, B., Qin, T., Wang, L., and Liu, T.-Y. Understanding and Improving Transformer From a Multi-Particle Dynamic System Point of View, June 2019. URL http://arxiv.org/abs/1906.02762. arXiv:1906.02762 [cs, stat].

Maheswaranathan, N., Williams, A., Golub, M. D., Ganguli, S., and Sussillo, D. Reverse engineering recurrent networks for sentiment classification reveals line attractor dynamics, December 2019. URL http://arxiv.org/abs/1906.10720. arXiv:1906.10720.

Olshausen, B. A. and Field, D. J. Sparse coding of sensory inputs. *Current opinion in neurobiology*, 14(4):481–487, 2004. Publisher: Elsevier.

Plaut, D. C. and McClelland, J. L. Locating object knowledge in the brain: Comment on Bowers's (2009) attempt to revive the grandmother cell hypothesis. *Psychological Review*, 2010. Publisher: American Psychological Association.

Shenoy, K. V., Sahani, M., and Churchland, M. M. Cortical Control of Arm Movements: A Dynamical Systems Perspective. *Annual Review of Neuroscience*, 36(1):337–359, July 2013. ISSN 0147-006X, 1545-4126. doi: 10.1146/annurev-neuro-062111-150509. URL https://www.annualreviews.org/doi/10.1146/annurev-neuro-062111-150509.

Singh, A. K., Moskovitz, T., Hill, F., Chan, S. C. Y., and Saxe, A. M. What needs to go right for an induction head? A mechanistic study of in-context learning circuits and their formation, April 2024. URL http://arxiv.org/abs/2404.07129. arXiv:2404.07129 [cs].

Vyas, S., Golub, M. D., Sussillo, D., and Shenoy, K. V. Computation Through Neural Population Dynamics. *Annual Review of Neuroscience*, 43